

RaceDay API Endpoint Plan

Purpose

This document defines the planned API endpoints for the RaceDay event management system. The API will allow users to register and log in, manage their profiles, browse and manage events, manage event categories, enrol in events, and view or record results.

The endpoints are planned for implementation in Part 2 using ASP.NET Core Web API.

User Roles

Organiser: Can create and manage events, categories, routes, event images, enrolments, and results.

Participant: Can manage their own profile, browse events, enrol in events, cancel their own enrolments, and view their results.

API Endpoints

HTTP method	Route	Description	Role required	Request body	Expected response
POST	/api/auth/register	Register a new user as an Organiser or Participant.	None – Public	{ email, password, firstName, lastName, role }	201 Created – user details without password; 400 Bad Request for validation errors; 409 Conflict if email already exists.
POST	/api/auth/login	Authenticate a user and provide a JWT token.	None – Public	{ email, password }	200 OK – token and user details; 401 Unauthorized for invalid credentials.
GET	/api/users/me	Retrieve the profile of the currently logged-in user.	Any logged-in user	None	200 OK – user profile.

HTTP method	Route	Description	Role required	Request body	Expected response
PUT	/api/users/me	Update the current user's profile information.	Any logged-in user	{ firstName, lastName, ... }	200 OK – updated profile.
GET	/api/events	Display a list of upcoming events for users to browse.	Public / Any	Optional query parameters	200 OK – list of events.
GET	/api/events/{id}	Retrieve detailed information about one event, including categories and route information.	Public / Any	None	200 OK – event details; 404 Not Found if the event does not exist.
POST	/api/events	Create a new race or sporting event.	Organiser	{ title, description, eventDate, location, ... }	201 Created ; 400 Bad Request ; 403 Forbidden .
PUT	/api/events/{id}	Update an event belonging to the organiser.	Organiser – Owner	Event information	200 OK ; 403 Forbidden ; 404 Not Found .
DELETE	/api/events/{id}	Delete an event belonging to the organiser.	Organiser – Owner	None	204 No Content ; 403 Forbidden ; 404 Not Found .
GET	/api/events/{eventId}/categories	Retrieve all categories belonging to an event.	Public / Any	None	200 OK – list of categories.
POST	/api/events/{eventId}/categories	Add a new category to an event.	Organiser – Owner	{ name, distanceKm, maxParticipants }	201 Created ; 400 Bad Request ; 403 Forbidden .

HTTP method	Route	Description	Role required	Request body	Expected response
PUT	/api/categories/{id}	Update an existing event category.	Organiser	Category information	200 OK ; 403 Forbidden ; 404 Not Found .
DELETE	/api/categories/{id}	Delete an event category.	Organiser	None	204 No Content ; 403 Forbidden ; 404 Not Found .
POST	/api/enrolments	Allow a participant to enrol in an event category.	Participant	{ categoryId }	201 Created ; 400 Bad Request if full ; 409 Conflict if already enrolled.
GET	/api/enrolments/my	Display the current participant's enrolments.	Participant	None	200 OK – list of enrolments.
GET	/api/events/{eventId}/enrolments	Allow an organiser to view enrolments for their event.	Organiser	None	200 OK – list of enrolments.
DELETE	/api/enrolments/{id}	Cancel an enrolment belonging to the current participant.	Participant – Owner	None	204 No Content ; 403 Forbidden if not the owner.
POST	/api/results	Record the result of a participant in an event.	Organiser	{ enrolmentId, finishTime, position, notes }	201 Created ; 409 Conflict if a result already exists.

HTTP method	Route	Description	Role required	Request body	Expected response
GET	/api/results/my	Allow a participant to view their own results and performance history.	Participant	None	200 OK – list of results.
GET	/api/events/{eventId}/results	Display results for a specific event.	Public / Any	None	200 OK – list of event results.
GET	/api/events/{eventId}/route	Retrieve route information for an event.	Public / Any	None	200 OK – route information; 404 Not Found if no route exists.
POST	/api/events/{eventId}/images	Add event image metadata. Images can later be stored using Azure Blob Storage.	Organiser	{ url, caption }	201 Created; 400 Bad Request; 403 Forbidden.

Endpoint Groups

The endpoints cover all required functional areas:

- **Authentication:** Registration and login.
- **User Profile:** Viewing and updating the current user's profile.
- **Events:** Creating, viewing, updating and deleting events.
- **Categories:** Creating, viewing, updating and deleting event categories.
- **Event Enrolments:** Registering participants, viewing enrolments and cancelling enrolments.
- **Results:** Recording and viewing race results.
- **Routes:** Viewing route information for an event.
- **Event Images:** Managing event image metadata for future Azure Blob Storage integration.

Security and Access

Role-based access will be used when the API is implemented. Public endpoints will not require authentication. Organiser endpoints will require an authenticated Organiser, while participant-specific endpoints will require an authenticated Participant.

Ownership checks will also be applied so that an organiser can only modify their own events and a participant can only manage their own enrolments and profile.